Lesson Planning

Complete Lesson Plan Book

Class: Class 7

Subject: Mathematics

Medium: English

Academic Year 2026-27

This book is prepared as a teaching guide for teachers.

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Unit 1: Sets

Learning Objectives

• Define a set and describe it by listing and set-builder methods.

• Identify finite, infinite, empty and singleton sets.

• Perform union, intersection and difference of two sets.

• Draw and read simple Venn diagrams.

Key Concepts

• A set is a well-defined collection of distinct objects.

• Elements, subset and equal sets; notation like $3 \in A$ and $5 \notin A$.

• Universal set and empty set ($\varnothing$ or $\{\}$).

• Union ($A \cup B$), intersection ($A \cap B$) and difference ($A - B$).

• Venn diagrams show relationships between sets visually.

Teaching Methodology (Step by Step)

1. Recall prior knowledge: ask students to name collections like vowels in the alphabet.
2. Introduce the definition of a set and write examples on the board.
3. Explain the two methods of describing sets with solved examples.
4. Demonstrate union, intersection and difference using shaded Venn diagrams.
5. Guide students through two board exercises, then let them solve one in pairs.

Classroom Activities

• Students sort classroom objects into labelled sets and list their elements.

• Draw Venn diagrams for the sets of even and odd numbers from 1 to 20.

Assessment Questions

1. List the set of first five prime numbers in both description methods.
2. If $A = \{1,2,3,4\}$ and $B = \{3,4,5,6\}$, find $A \cup B$, $A \cap B$ and $A - B$.
3. Draw a Venn diagram for the sets of multiples of 2 and 3 up to 18.

Homework

• Write 10 examples of sets from daily life and identify which are finite or infinite.

Time Allocation

Introduction 5 min, concept development 20 min, practice 10 min, wrap-up 5 min.

Unit 2: Algebraic Expressions

Learning Objectives

- Identify variables, constants, terms and coefficients.
- Simplify algebraic expressions by combining like terms.
- Add and subtract algebraic expressions.
- Evaluate expressions for given values of variables.

Key Concepts

- A variable is a symbol (like x) that can take different values.
- Like terms have the same variable parts; unlike terms cannot be combined directly.
- The coefficient is the number multiplied by the variable.
- Addition and subtraction are done term by term.
- Substitution replaces variables with numbers to evaluate an expression.

Teaching Methodology (Step by Step)

1. Revise arithmetic operations with a quick mental-maths drill.
2. Introduce algebraic vocabulary with a labelled diagram of $5x^2 + 3x - 7$.
3. Model combining like terms on the board step by step.
4. Show vertical addition and subtraction of expressions.
5. Students evaluate expressions on mini whiteboards for teacher checking.

Classroom Activities

- Card matching game: match expression cards with their simplified forms.
- Pair quiz where one student gives values of x and the other evaluates $4x + 9$.

Assessment Questions

1. Identify the terms and coefficients in $7a - 2b + 5$.
2. Simplify: $(6x + 3y - 2) + (4x - 5y + 8)$.
3. Evaluate $3x^2 - 2x + 1$ when $x = 4$.

Homework

- Simplify five expressions and evaluate two of them for $x = -2$.

Time Allocation

Drill 5 min, concept teaching 20 min, board practice 10 min, summary 5 min.

Unit 3: Linear Equations

Learning Objectives

• Define a linear equation in one variable.

• Solve equations of the form ax + b = c.

• Verify solutions by substitution.

• Translate simple word problems into equations.

Key Concepts

• An equation is a statement that two expressions are equal.

• Balance method: do the same operation on both sides to keep the equation true.

• Solution of an equation is the value of the variable that satisfies it.

• Inverse operations undo each other (addition undoes subtraction).

• Word statements like 'five more than a number' become x + 5.

Teaching Methodology (Step by Step)

1. Use a balance-scale picture to show what an equation means.
2. Demonstrate solving x + 7 = 15 by removing 7 from both sides.
3. Solve 3x − 4 = 11 showing two steps clearly.
4. Model translating a word problem: 'twice a number plus 6 is 20'.
5. Students solve a graded set: one-step, two-step, then one word problem.

Classroom Activities

• Balance-scale role play: volunteers hold number cards on each side.

• Equation relay race where teams solve equations on the board.

Assessment Questions

1. Solve: 2x + 9 = 25 and 5x − 7 = 3x + 9.
2. A number multiplied by 4 and then reduced by 5 gives 27. Find the number.
3. Check whether x = 6 is the solution of 7x − 5 = 37.

Homework

• Solve eight equations and make one word problem of your own with its solution.

Time Allocation

Motivation 5 min, teaching 20 min, practice 10 min, homework briefing 5 min.

Unit 4: Ratio and Proportion

Learning Objectives

- Define ratio and write it in simplest form.
- Solve problems of direct and inverse proportion.
- Divide a quantity in a given ratio.
- Apply ratio to unitary-method problems.

Key Concepts

- A ratio compares two quantities of the same kind, e.g. 3 : 4.
- Ratios must be in the same units before comparing.
- In direct proportion, when one quantity rises the other rises too.
- In inverse proportion, when one rises the other falls.
- Unitary method finds the value of one unit first, then the required units.

Teaching Methodology (Step by Step)

1. Ask students to compare the number of boys and girls in class as a ratio.
2. Teach simplifying ratios with common factors.
3. Introduce direct proportion using the example of notebooks and price.
4. Introduce inverse proportion using workers and days to finish a job.
5. Work a division-in-ratio problem on the board and assign practice.

Classroom Activities

- Mix coloured water in given ratios and describe the mixtures.
- Group task: divide 120 marbles among three friends in the ratio 2 : 3 : 5.

Assessment Questions

1. Simplify the ratio 45 : 75.
2. If 8 pens cost Rs 200, what is the cost of 13 pens?
3. Divide Rs 900 between A and B in the ratio 4 : 5.

Homework

- Solve six ratio and proportion word problems from the worksheet.

Time Allocation

Warm-up 5 min, teaching 20 min, activity 10 min, review 5 min.

Unit 5: Geometry: Angles and Triangles

Learning Objectives

• Classify angles as acute, right, obtuse and straight.

• Identify complementary and supplementary angles.

• Classify triangles by sides and by angles.

• State and use the angle-sum property of a triangle.

Key Concepts

• Angles are measured in degrees; a full turn is 360°.

• Complementary angles add to 90°; supplementary angles add to 180°.

• Equilateral, isosceles and scalene triangles differ in their sides.

• Acute, right and obtuse triangles differ in their largest angle.

• The three interior angles of a triangle always add to 180°.

Teaching Methodology (Step by Step)

1. Revise point, line and ray with a quick board sketch.
2. Demonstrate measuring angles with a protractor.
3. Explain complementary and supplementary pairs with examples.
4. Show triangle types using paper cut-outs.
5. Verify the angle-sum property by tearing the corners of a paper triangle.

Classroom Activities

• Students measure angles around the classroom with a protractor.

• Paper-tearing activity to discover that triangle angles sum to 180°.

Assessment Questions

1. Find the complement of 37° and the supplement of 112°.
2. Two angles of a triangle are 55° and 68°. Find the third angle.
3. Classify a triangle with sides 5 cm, 5 cm and 8 cm.

Homework

• Draw each type of triangle and label its angles after measuring them.

Time Allocation

Revision 5 min, teaching 20 min, hands-on activity 10 min, quiz 5 min.

Unit 6: Perimeter and Area

Learning Objectives

• Recall formulas for perimeter of square, rectangle and triangle.

• Calculate the area of squares, rectangles and triangles.

• Find the circumference and area of a circle.

• Solve practical problems involving fencing and flooring.

Key Concepts

• Perimeter is the total distance around a shape.

• Area of a rectangle is length × width; of a triangle is ½ × base × height.

• Circumference of a circle is $2\pi r$ and area is πr^2.

• Use $\pi = 22/7$ or 3.14 as instructed.

• Real-life uses: fencing a field, painting a wall, laying tiles.

Teaching Methodology (Step by Step)

1. Revise perimeter by walking a finger around drawn shapes.
2. Derive area formulas with grid paper shading.
3. Introduce the circle formulas with a string-around-the-disc demo.
4. Solve a fencing problem and a tiling problem on the board.
5. Students compute areas of classroom floor tiles in pairs.

Classroom Activities

• Measure the classroom and estimate the cost of tiling it.

• Cut a paper circle into sectors and rearrange them into a near-rectangle to see πr^2.

Assessment Questions

1. Find the area of a triangle with base 12 cm and height 9 cm.
2. A circular garden has radius 7 m. Find its circumference and area.
3. A rectangular plot 40 m by 25 m is to be fenced. Find the length of wire needed.

Homework

• Solve five problems on perimeter and area including one circle problem.

Time Allocation

Revision 5 min, teaching 20 min, activity 10 min, assessment 5 min.

Unit 7: Data Handling

Learning Objectives

• Organise raw data into a frequency table.

• Draw and read bar graphs and pie charts.

• Calculate the mean of a data set.

• Find the median and mode of simple data.

Key Concepts

• Data is information collected as numbers or facts.

• Tally marks help count frequencies quickly.

• A bar graph uses bars of equal width to compare quantities.

• A pie chart shows parts of a whole as sectors of a circle.

• Mean is the average; median is the middle value; mode is the most frequent value.

Teaching Methodology (Step by Step)

1. Collect class data: favourite fruit of each student.
2. Build a frequency table using tally marks on the board.
3. Draw a bar graph of the data step by step.
4. Show how to convert frequencies into a pie chart.
5. Compute mean, median and mode of the class heights sample.

Classroom Activities

• Survey the class and present the results as a bar graph.

• Group project: draw a pie chart of how students travel to school.

Assessment Questions

1. Find the mean, median and mode of 4, 7, 9, 7, 12, 7, 15.
2. Draw a bar graph for the data: cricket 12, football 8, hockey 5.
3. What is the mode of the data set above?

Homework

• Collect the ages of family members and find their mean and median.

Time Allocation

Data collection 5 min, teaching 20 min, drawing 10 min, recap 5 min.

Unit 8: Rational Numbers

Learning Objectives

• Define rational numbers and give examples.

• Represent rational numbers on the number line.

• Compare rational numbers by converting to common denominators.

• Add, subtract, multiply and divide rational numbers.

Key Concepts

• A rational number can be written as p/q where q ≠ 0.

• Integers, fractions and terminating decimals are all rational.

• Standard form has a positive denominator with no common factors.

• Addition and subtraction need a common denominator (LCM).

• Division by a fraction means multiplying by its reciprocal.

Teaching Methodology (Step by Step)

1. Revise integers and fractions with a number-line walk.
2. Introduce the definition with everyday examples like 3/4 of a pizza.
3. Show standard form and equivalent fractions.
4. Demonstrate addition and subtraction with common denominators.
5. Teach multiplication and division with worked board examples.

Classroom Activities

• Number-line relay: students place rational numbers on a floor number line.

• Fraction-card game for multiplying and dividing pairs of fractions.

Assessment Questions

1. Write the standard form of −18/24.
2. Compute 5/6 + (−7/12) and 4/9 ÷ (−2/3).
3. Arrange −3/5, 1/2 and −1/4 in ascending order.

Homework

• Solve ten mixed-operation problems on rational numbers.

Time Allocation

Revision 5 min, concept teaching 20 min, practice 10 min, wrap-up 5 min.